

Reliability-Calibrated Online Mirror Descent

What survives sparse rewards — and what breaks under shared parameters

TEAM PRISM · RL-GROUP 1

HUANG YUXUAN

Project lead & final report

LIU YIPU

Theory

XIE HAOXIANG

Environments & estimators

ZHONG YIFAN

Algorithms & infrastructure

LI YUFEI

Experiments & analysis

A five-person optimization study with cross-review of evidence and claims.

One terminal bit must supervise a long trajectory

SPARSE SIGNAL

$$R(\tau) \in \{0,1\}$$

The verifier scores the completed trajectory, not each intermediate decision.

GROUP BASELINE

$$A_i = r_i - \bar{r}$$

Group-relative updates compare sampled outcomes without training a separate critic.

BROADCAST

$$A_i \rightarrow \text{every step}$$

The same sequence-level signal can move pivotal and routine decisions together.

Our boundary: this is an associative credit-allocation study — not a causal credit proof.

RC-OMD separates direction from permitted movement

credit estimate $\hat{c}_{t,k}$

chooses WHERE



reliability $q_{t,k}$

chooses HOW FAR



$\eta_{t,k} = \eta_0 f(q_{t,k})$

local KL / step-size budget

1 ONLINE RELIABILITY

A persistent standardized score updates after each grouped rollout.

2 LOCAL GEOMETRY

Reliable evidence permits larger movement; weak evidence stays conservative.

3 EVALUATION

Known distractor positions are used only to measure wasted KL in controlled tasks.

Tabular, factorized policy only in the positive study.

Pre-registered OOD result: 4 of 4 scenarios pass

GO

OOD scenario (n = 10 seeds)	Uniform AUC	Online AUC	AUC Δ	Distractor KL ratio
Dense 2-of-6 · tiny group	0.992182	0.990865	-0.001317	0.248
Needle 5-of-5 · long horizon	0.980026	0.980782	+0.000755	0.239
Threshold 3-of-5 · small group	0.992195	0.990567	-0.001628	0.189
Threshold 4-of-6 · 3 actions	0.989446	0.987688	-0.001758	0.199

 ≤ 0.0018

absolute AUC difference

18.9%–24.8%

of Uniform distractor KL

6.9%–10.7%

runtime overhead · all < 1.5×

Shared parameters break the local-control mechanism

1 / 3 PASS

Distractor-KL ratio (Online / Uniform)



AUC conditions passed in all 3; failure came entirely from coupled distractor KL.

Questions for the professor

01

GEOMETRY

Should reliability enter a Fisher / natural-gradient metric?

02

PROJECTION

Can we fit pivotal OMD targets while constraining low-reliability KL?

03

NEXT TEST

Which nonlinear benchmark should precede any neural or LLM claim?

NO neural-network or LLM effectiveness claim.